

Electrical Engineering:

Problem Statement No.16 – Smart Home Energy Advisor Agent

The Challenge

Rising electricity demand and energy costs make it difficult for households to understand how their daily habits impact power consumption. Most users only receive monthly bills, which provide limited insight into appliance-level energy usage. As a result, people often waste electricity unknowingly, highlighting the need for an AI-powered system that analyzes usage patterns and provides actionable energy-saving recommendations.

The Objective

Build an AI-powered Smart Home Energy Advisor Agent that analyzes household electricity consumption data and provides intelligent recommendations to optimize energy usage.

1. **Energy Consumption Analysis** – Analyze household electricity usage patterns from smart meters or appliance usage logs.
2. **Personalized Energy Recommendations** – Suggest energy-saving actions based on the user's consumption behavior and daily routines.
3. **Appliance Usage Insights** – Identify appliances that consume the most electricity and recommend optimal usage strategies.
4. **Electricity Bill Estimation** – Predict monthly electricity bills based on current usage patterns and tariff structures.
5. **Energy Efficiency Suggestions** – Recommend simple practices such as reducing peak-hour usage, scheduling heavy appliances, or upgrading to energy-efficient devices.

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- Lang Flow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with Lang Chain + RAG + IBM Granite Models